



Own the AI Agent

Mint an NFT with its own wallet. Approve a plan once.
Keepers execute every step on-chain.

The Problem

Centralized

Agents run on
servers you don't
control

No ownership

No on-chain limits
or verifiable execu-
tion

Manual ops

Babysit infra, click
every step

You rent the agent. You don't own it.

The Solution

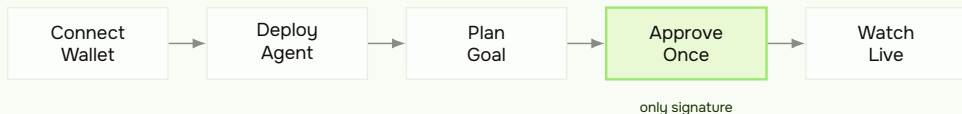
- ERC-721 NFT owns an ERC-6551 wallet on Somnia
- Fund once, set caps, approve one plan
- Keepers + validators dispatch, rate, and pay on-chain
- No user-operated server. No cron jobs.

One signature

1×

The agent runs itself

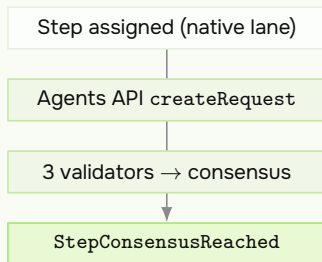
How It Works



Approve Once is the only MetaMask popup in the entire demo

Somnia Native Agentation

- Native steps route through the Somnia Agents API
- Orchestrator calls `createRequest` with ABI-encoded payloads
- 3-validator subcommittee reaches consensus on-chain
- `StepConsensusReached` – verifiable validator receipts
- Relay keeper bridges results back to `AgentOrchestrator`



Native catalog (configIds 0-5): janice · web-intel · somnia-oracle · analysis-bot · reporter-bot · executor-bot

Somnia Primitives in Twiin

ERC-6551

NFT owns a token-bound account – the agent's on-chain wallet. One `deployTwiin` tx mints, funds, and seeds policy.

Agents API

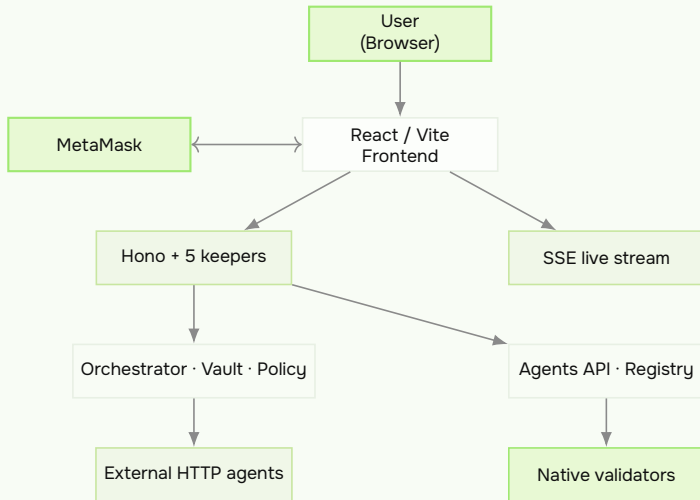
Native lane dispatches to Somnia base agents via on-chain requests. Validator consensus – not a centralized cron.

Reactivity

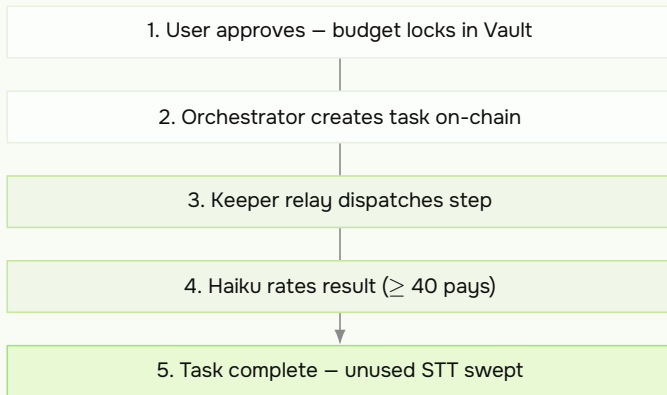
`subscribePull` + `pullForRefresh` on the 6551 account. Chain-side oracle refresh without off-chain schedulers.

Built for Somnia Shannon testnet · chainId 50312 · STT gas

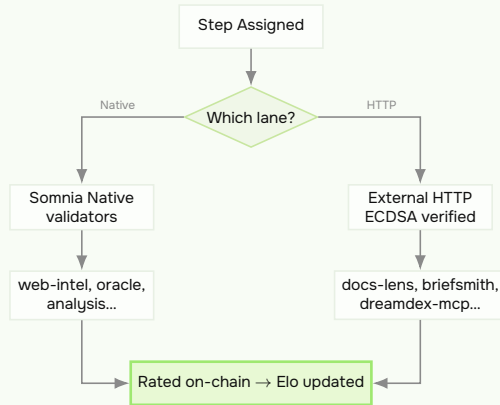
System Architecture



Task Lifecycle

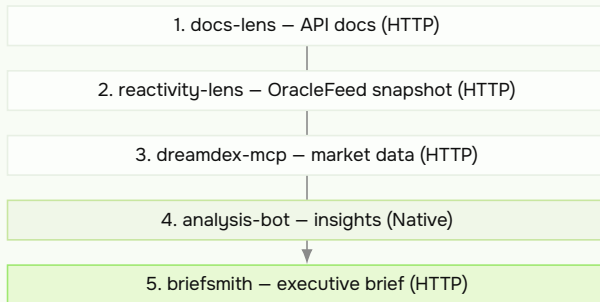


Two-Lane Registry



Live Demo Pipeline

Goal: How healthy is the Somnia ecosystem?



One signature starts the entire pipeline

Built & Shipped

94+

Contract tests

6

Native agents

5

Keeper bots

7

External agents

Full stack

- 11 Solidity contracts on Somnia
- React console + marketplace UI
- Claude Haiku planner & rater

Live

- twiin-frontend.vercel.app
- github.com/wurli-sh/twiin
- Somnia Shannon · chainId 50312

Thank You

`github.com/wurli-sh/twiin`

`twiin-frontend.vercel.app`